Conflict-Aware RAG Routing: Balancing Cost and Accuracy via Context Contradiction

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Abstract—Adaptive model routing can reduce the cost of retrieval-augmented generation (RAG), but the routing signal must identify when a small language model (SLM) is likely to fail. We study context conflict, the contradiction among retrieved passages, as one such signal. A five-feature Random Forest combines conflict with query length, retrieved-context length, chunk overlap, and Reciprocal Rank Fusion (RRF) confidence. We evaluate the pipeline on 3,000 multi-hop questions from MuSiQue, HotpotQA, and 2WikiMultiHopQA using Qwen2.5-1.5B-Instruct locally and DeepSeek-V4-Flash through an API. Nested five-fold calibration selects budget cutoffs from inner out-of-fold rather than in-sample training scores. At an approximately 65% LLM-call rate, the hybrid router saves about 35% of API calls while retaining 83.2–92.2% of the Always-LLM F1. Its advantage over RRF alone is dataset-dependent: at the 50% budget it changes F1 by -1.08, -0.03, and $+1.48$ points on MuSiQue, HotpotQA, and 2Wiki, respectively; only the 2Wiki interval excludes zero. At 30%, it offers no consistent advantage. An exploratory high-RRF/high-conflict subgroup analysis finds larger hybrid-versus-RRF gains on HotpotQA and 2Wiki. However, a direct additive conflict override at $+2/5/10$ usage points does not outperform equal-cost random escalation. Thus, contradiction can be complementary on some retrieval distributions, but a simple conflict override is not generally justified.

Index Terms—retrieval-augmented generation, dynamic model routing, context contradiction, natural language inference, multi-hop question answering

I. Introduction

Retrieval-Augmented Generation (RAG) supplies a language model with external evidence and can improve knowledge-intensive question answering [1]. Calling a large API model for every question, however, incurs monetary and latency costs. Adaptive routing instead sends easier instances to a local SLM and reserves the stronger LLM for instances on which escalation is valuable. Related systems adapt retrieval depth or model use from question complexity, confidence, or self-reflection [2], [3].

Most retrieval-aware gates use relevance or rank-based confidence. These scores do not directly describe whether the retrieved passages disagree. Irrelevant or distracting context can impair reasoning [4]; this sensitivity may be especially consequential for compact models. We therefore examine whether contradiction among the retrieved passages helps predict when the API LLM will outperform the local SLM.

This paper makes three contributions. First, it specifies a reproducible context-conflict score based on bidirectional natural language inference (NLI). Second, it compares fixed heuristics, a four-feature Random Forest, and a five-feature conflict-aware Random Forest under nested, leakage-resistant budget calibration. Third, it reports practical effect sizes, confidence intervals, realized LLM usage, and failure cases across three multi-hop datasets. The experiment supports a narrower conclusion than a universal win: routing can approach the Always-LLM endpoint with fewer calls, while the incremental value of conflict varies by dataset and budget.

II. Related Work

Lewis et al. introduced RAG for knowledge-intensive NLP [1]. Adaptive-RAG learns to select retrieval strategies according to question complexity [2], while Self-RAG learns reflection tokens for retrieval and critique [3]. Our objective differs: retrieval is fixed, and the decision is which generator should answer.

Hybrid retrieval commonly combines sparse and dense ranks. We use Reciprocal Rank Fusion (RRF), a robust rank-combination method [5]. The conflict feature is computed with DeBERTa-v3, an NLI-capable encoder architecture [6]. Unlike an RRF score, which describes how strongly evidence was retrieved, NLI contradiction describes relations within the retrieved evidence. The experiment tests whether these signals are complementary rather than assuming independence.

The evaluation uses MuSiQue [7], HotpotQA [8], and 2WikiMultiHopQA [9]. All require evidence integration across multiple facts, making them suitable for studying retrieval noise and model escalation.

III. Method

A. Retrieval and Generation Context

For each question, the supplied corpus is segmented into 1,024-character parent chunks and 256-character child chunks with no overlap; document titles are prepended. BM25 and all-MiniLM-L6-v2 each return 20 children. Their ranks are fused with RRF using $k = 60$. Each parent receives the maximum score of its retrieved children. The five highest-ranked parents form the generation context; the top three are used for conflict scoring. The maximum

parent RRF score is the retrieval-confidence feature. Chunk overlap is the lexical overlap between the question and the retrieved context.

B. Context Conflict

Let $P = \{p_1, p_2, p_3\}$ be the top three parents. For each of the three unordered pairs, the cross-encoder/nli-deberta-v3-small model scores both input orders. We average the two contradiction probabilities as $q_{ij} = [S(p_i, p_j) + S(p_j, p_i)]/2$ to remove direction dependence. With $Q = \{q_{12}, q_{13}, q_{23}\}$, the aggregate is

$$C = 0.7 \max_{q \in Q} q + 0.3 \,\mathrm{mean}_{q \in Q}\, q. \tag{1}$$

The maximum captures one sharp pairwise disagreement and the mean broader inconsistency. The 0.7/0.3 weighting is an a priori heuristic and was not tuned on the evaluation data.

C. Routers and Learning Target

The endpoint baselines are Always-SLM and Always-LLM. A true oracle, which chooses the answer having the larger per-question F1, is reported only as an unattainable upper reference. Feasible heuristic baselines are low-RRF routing and high-conflict routing. The learned four-feature router uses query length, context length, RRF confidence, and chunk overlap. The proposed five-feature router adds C.

Both learned routers are Random Forest classifiers with 500 trees, the Gini split criterion, unrestricted depth, minimum leaf size 2, square-root feature subsampling, balanced class weights, and seed 9,302,026. The binary target is one exactly when the LLM answer obtains greater token F1 than the SLM answer. The forest score ranks queries for escalation; it is not interpreted as a calibrated probability of correctness.

D. Nested Budget Calibration

We use fixed five-fold outer cross-validation. Within each outer training fold, five inner folds generate out-of-fold scores. For each target budget $b \in \{0.30, 0.50, 0.65\}$, the cutoff is selected only from these inner out-of-fold scores and then applied to the outer test fold. Ties are resolved by a label-free SHA-256 ordering. This avoids selecting a threshold from predictions on the outer test instances. We report the concatenated outer-fold predictions and their realized LLM usage, which can differ slightly from b because folds are finite and scores can tie.

IV. Experimental Setup

A. Data and Reproducibility

We evaluate 1,000 validation questions from each dataset. Sampling is deterministic and recorded by question identifiers and frozen input hashes. Retrieval and conflict features contain 3,000 unique rows. Dataset answer aliases are used when available. Token-level F1 follows normalized QA evaluation: lowercase text, remove punctuation and articles, normalize whitespace, and compute token overlap.

TABLE I
Endpoint F1 and Retrieval Quality.

Dataset	SLM	LLM	Hit@5	MRR@5
MuSiQue	13.01	24.15	95.8	83.55
HotpotQA	23.85	48.53	77.2	53.41
2Wiki	16.40	25.45	62.9	37.04

B. Models and Decoding

The local endpoint is Qwen2.5-1.5B-Instruct [10], pinned to model revision 989aa798. The API endpoint is DeepSeek-V4-Flash [11] in non-thinking mode. Both receive the same question and top-five retrieved parents and are instructed to answer briefly using the context. Generation uses temperature 0.1, top-p 0.95, and at most 50 new tokens. The API run recorded one production backend fingerprint for all 3,000 rows. Because an API-side seed was unavailable, reproducibility relies on frozen outputs, identifiers, request settings, and hashes rather than bitwise regeneration.

C. Uncertainty and Comparisons

Mean F1 is accompanied by a 95% percentile bootstrap interval using 1,000 query-level resamples. For the primary comparison at 50%, we bootstrap paired per-question differences between the hybrid and RRF routers and report paired Cohen's d_z. We emphasize the interval and magnitude before the two-sided bootstrap p-value. Raw two-sided bootstrap p-values are adjusted by the Holm procedure across the three datasets; confidence intervals remain pointwise.

V. Results

A. Endpoint and Retrieval Performance

Table I establishes the gap the router attempts to close. DeepSeek-V4-Flash exceeds the local SLM on all three datasets, but absolute F1 varies substantially. Retrieval is strongest on MuSiQue and weakest on 2Wiki. The oracle sometimes uses fewer than half of the LLM calls because the LLM is not superior on every instance; it is an upper reference, not a deployable policy.

B. Cost–Accuracy Trade-Off

Table II reports F1 with realized LLM usage in parentheses. At the 65% target, the hybrid uses 65.0%, 64.2%, and 63.9% of LLM calls on MuSiQue, HotpotQA, and 2Wiki. Its F1 retains respectively 83.8%, 83.2%, and 92.2% of Always-LLM F1, saving approximately 35% of calls. This is the principal deployment result.

The hybrid's incremental benefit over simpler routing is conditional. At 30%, it trails RRF on MuSiQue, is effectively tied on HotpotQA, and is higher on 2Wiki. At

50%, its largest improvement is on 2Wiki; on MuSiQue it is lower than RRF. Figure 1 shows that no single router dominates across all datasets and budgets.

C. Practical Effect at the 50% Budget

Relative to RRF, the hybrid changes F1 by -1.08 points on MuSiQue (95% paired bootstrap CI $[-2.38, 0.24]$, $d_z = -0.048$, raw $p = 0.124$, Holm-adjusted $p = 0.248$), -0.03 on HotpotQA ($[-1.91, 1.86]$, $d_z = -0.001$, raw and adjusted $p = 0.979$), and $+1.48$ on 2Wiki ($[0.08, 2.87]$, $d_z = 0.066$, raw $p = 0.040$, adjusted $p = 0.120$). The pointwise 2Wiki interval excludes zero, but its test does not remain significant after correction. Its small standardized effect reflects the large dispersion of paired per-query differences relative to the aggregate mean gain. The other intervals include zero. These results do not support a claim of consistent superiority across datasets.

D. What the Forest Learns

Table III reports mean Gini importance across the five outer forests, with standard deviations across folds. Context conflict and lexical overlap are the largest features on HotpotQA and 2Wiki, whereas context length and overlap are largest on MuSiQue. Gini importance describes feature use inside the fitted forest; it does not establish causal contribution.

Conflict and RRF are related only weakly: Spearman $\rho = 0.090$ on MuSiQue, 0.064 on HotpotQA, and 0.170 on 2Wiki. With 1,000 instances, each correlation is statistically detectable ($p = .0045$, .0417, and $< 10^{-7}$), so "orthogonal" would be too strong. The small coefficients instead support the narrower interpretation that conflict contains information not captured by RRF alone.

E. Where Conflict Changes Routing

We conducted an exploratory analysis to identify a failure mode rather than select a favorable conflict threshold. Within each dataset, label-free SHA-256 tie breaking partitions conflict into exact rank tertiles. Mean LLM-minus-SLM F1 does not increase monotonically from low to high conflict: the three values are 11.21, 10.35, and 11.84 points on MuSiQue; 24.12, 26.10, and 23.83 on HotpotQA; and 10.81, 9.52, and 6.82 on 2Wiki. Conflict alone is therefore not a general scalar measure of endpoint advantage (Fig. 2a).

The more informative case is disagreement between signals. We define a subgroup without answer labels as the intersection of the top third of RRF confidence and the top third of conflict. At the 50% target, RRF leaves these high-confidence queries on the SLM, whereas the hybrid can override that choice. Within this subgroup, hybrid-minus-RRF F1 is $+14.35$ on HotpotQA ($n = 114$, 95% paired bootstrap CI [8.85, 20.01], $d_z = .453$) and $+6.50$ on 2Wiki ($n = 131$, [2.55, 10.76], $d_z = .266$); both Holm-adjusted $p = .006$ across the three datasets. MuSiQue improves by $+1.70$ ($n = 123$, $[-0.01, 3.85]$, $d_z = .148$, adjusted $p = .058$). Thus, conflict is most defensible as an exception signal when retrieval appears confident, not as a universal difficulty score.

The subgroup gain is not equivalent to global dominance. At 50%, hybrid escalations that RRF omits contribute $+5.86$, $+2.75$, and $+1.71$ F1 points over all HotpotQA, 2Wiki, and MuSiQue queries, but hybrid de-escalations contribute -5.89, -1.27, and -2.79 points, respectively. The two effects cancel on HotpotQA, remain positive on 2Wiki, and become negative on MuSiQue.

F. Additive Conflict Override

We next test a deployment-oriented policy without new endpoint calls. Starting from each out-of-fold RRF route, we preserve every RRF escalation and add exactly 20, 50, or 100 LLM calls per 1,000 questions ($+2/5/10$ absolute usage points), choosing the highest-conflict questions from the RRF-to-SLM pool. A control adds the same number uniformly at random from the identical pool. Table IV reports the 50% target; conclusions are unchanged at the 30% and 65% targets.

Additional calls generally improve on RRF, especially on HotpotQA, but conflict does not rank them better than chance: all 27 conflict-versus-random 95% paired-bootstrap intervals include zero and all Holm-adjusted p-values equal 1.0. No override-versus-RRF test remains below .05 after correction (minimum adjusted $p = .0539$). The high-RRF/high-conflict result therefore diagnoses fitted-hybrid behavior inside a subgroup; it does not validate a global top-conflict override.

G. Ablation, Dense Budgets, and Transfer

A paired five-feature-minus-four-feature ablation confirms that Gini importance does not imply incremental F1. None of nine dataset/budget comparisons survives Holm correction. The largest raw effect is $+1.12$ points on 2Wiki at 30% (95% CI [0.11, 2.18], raw $p = .028$, adjusted $p = .252$); the other intervals include zero.

To avoid conclusions tied to three operating points, we repeat nested calibration from 0 to 100% in 2.5-point steps and integrate F1 over realized usage. Relative to RRF, hybrid AURC changes by -0.39, $+0.45$, and $+0.63$ F1 points on MuSiQue, HotpotQA, and 2Wiki; hybrid exceeds RRF at 2/41, 28/41, and 39/41 grid points, respectively. Relative to the four-feature RF, the AURC changes are -0.28, -0.06, and $+0.31$. These descriptive results reinforce dataset dependence rather than universal superiority.

Finally, leave-one-dataset-out transfer does not support a dataset-general conflict feature. For example, adding conflict changes F1 by -2.62 on held-out HotpotQA at the 30% target (95% CI $[-4.00, -1.20]$) and by -1.27 on held-out 2Wiki at 50% ($[-2.39, -0.26]$), with material usage drift. Target labels are used only for evaluation. The conflict-aware override therefore requires in-domain calibration.

TABLE II
Nested Five-Fold Routing Results: Token F1 with Realized LLM Usage (%) in Parentheses. Bold Marks the Best Feasible Method Within Each Dataset/Budget Block; Endpoints and Oracle Are References.

Target	Router	MuSiQue	HotpotQA	2Wiki
0%	Always SLM	13.01 (0.0)	23.85 (0.0)	16.40 (0.0)
100%	Always LLM	24.15 (100.0)	48.53 (100.0)	25.45 (100.0)
–	True oracle	27.84 (43.6)	51.00 (55.6)	30.45 (42.2)
30%	RRF confidence	16.74 (30.1)	31.22 (30.1)	19.17 (29.7)
	Conflict threshold	16.29 (29.8)	31.18 (30.2)	18.56 (30.1)
	4-feature RF	16.39 (27.5)	31.26 (28.8)	19.35 (30.2)
	5-feature hybrid	16.10 (28.2)	31.28 (28.9)	20.47 (31.9)
50%	RRF confidence	19.51 (50.0)	36.32 (50.0)	20.83 (49.8)
	Conflict threshold	18.99 (49.9)	36.63 (50.0)	20.35 (49.8)
	4-feature RF	18.86 (49.0)	36.00 (48.9)	21.99 (51.3)
	5-feature hybrid	18.43 (49.9)	36.29 (48.6)	22.30 (50.7)
65%	RRF confidence	20.85 (65.0)	39.72 (64.8)	22.43 (65.2)
	Conflict threshold	20.40 (64.8)	40.14 (64.8)	21.75 (64.6)
	4-feature RF	20.71 (64.6)	40.33 (63.6)	23.42 (64.8)
	5-feature hybrid	20.24 (65.0)	40.38 (64.2)	23.46 (63.9)

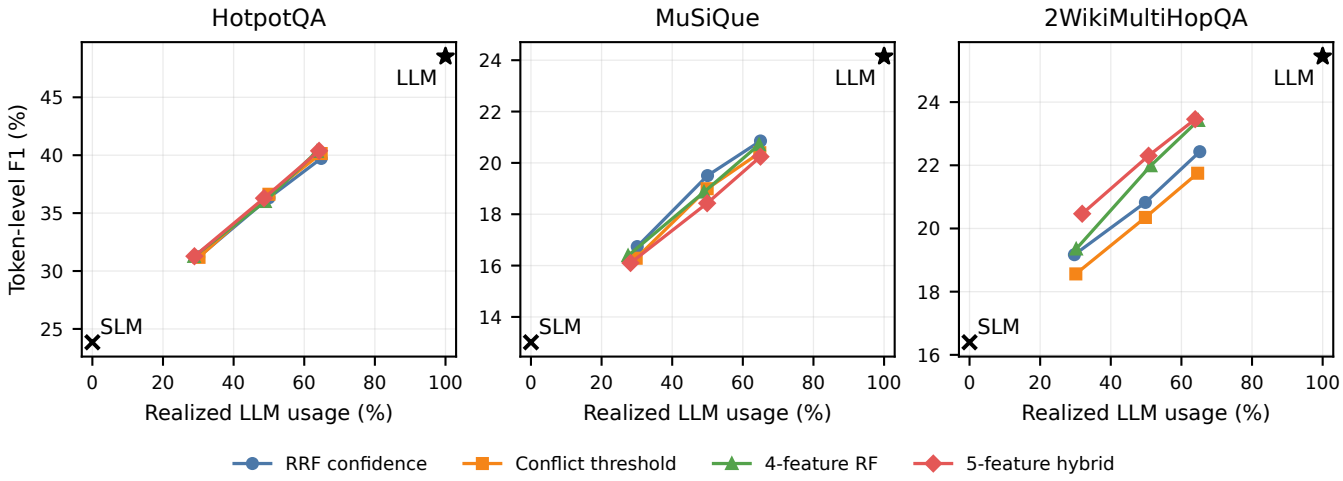


Fig. 1. F1 against realized LLM usage. Crosses and stars denote Always-SLM and Always-LLM; connected points correspond to 30%, 50%, and 65% target budgets.

VI. Limitations

Only one SLM, one API LLM, and 1,000 examples per dataset are evaluated. API behavior can change and was not seed-controlled. The 50-token cap truncated 160 of 3,000 API responses (5.3%); these responses have lower mean F1 on all three datasets. Mean API latency also contains a 330.96-second tail event, making latency comparisons unstable. We therefore focus on call rate, not a claim of guaranteed latency reduction.

The high-RRF/high-conflict subgroup analysis is exploratory and uses the same endpoint outputs as the main evaluation; it requires prospective validation on independent data.

The cost axis treats each LLM call equally and omits token-dependent API pricing, local hardware amortization, and the NLI scorer's compute. Pairwise NLI is quadratic in the number of passages, although only three passages are scored here. Leave-one-dataset-out evaluation shows material usage drift and no transferable conflict advantage; temporal calibration also remains open.

TABLE III
Five-Feature RF Gini Importance, Mean ± SD Across Outer Folds.

Feature	MuSiQue	HotpotQA	2Wiki
Query length	.157 ± .002	.150 ± .002	.136 ± .002
Context length	.227 ± .002	.238 ± .006	.251 ± .006
RRF confidence	.128 ± .004	.092 ± .001	.085 ± .003
Chunk overlap	.244 ± .005	.263 ± .006	.264 ± .009
Context conflict	.244 ± .006	.256 ± .002	.265 ± .004

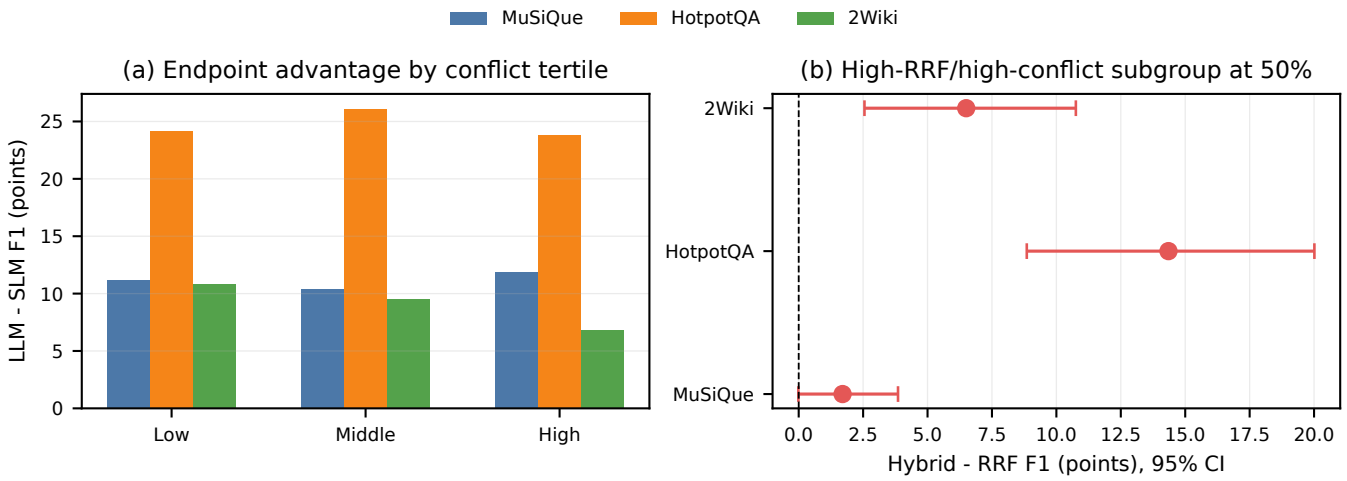


Fig. 2. Exploratory failure-mode analysis. (a) High conflict does not imply a monotonically larger LLM advantage. (b) Nevertheless, at the 50% target the hybrid recovers errors missed by RRF inside the label-free high-RRF/high-conflict subgroup on HotpotQA and 2Wiki.

TABLE IV
Additive Conflict Override at the 50% RRF Target. Differences Are F1 Points; Random Is the Exact Expected F1 of Uniform Extra Calls at Identical Cost.

Dataset	Extra calls	Realized usage	Override − RRF	Override − random
MuSiQue	+2/5/10%	52/55/60%	+0.10/ +0.35/ +0.75	−0.08/ −0.11/ −0.18
HotpotQA	+2/5/10%	52/55/60%	+0.57/ +0.88/ +2.56	+0.08/ −0.35/ +0.12
2Wiki	+2/5/10%	51.8/54.8/59.8%	+0.18/ +0.31/ +0.48	−0.00/ −0.15/ −0.44

VII. Conclusion

Context conflict is not a universal difficulty score: adding it to the RF does not yield a corrected-significant global ablation, transfer is poor, and a monotone top-conflict override does not beat equal-cost random escalation. Its defensible role is narrower: the fitted hybrid identifies a high-RRF/high-conflict failure regime on HotpotQA and 2Wiki, but turning that diagnosis into a deployable policy requires separate calibration. Across all queries, routing still provides a cost–quality trade-off, using roughly 64–65% of LLM calls at the 65% target while retaining 83–92% of Always-LLM F1. Future work should learn a non-monotone override gate on separate validation data, preserve beneficial RRF escalations, and test calibration under shift.

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